Relatedness and Structural Dependence

Greater Manchester (UKD3) <–> China (CN) | openalex | log RCA axis; dotted = RCA 1 & median reldens; dashed = log−linear fit

Pearson (log RCA) = 0.10 (p=0.525) Spearman = 0.18 (p=0.260) n = 39

Greater Manchester (UKD3)'s reliance on China (CN) (RCA)

Mobile Networks (5G/6G)
Internet of Things (IoT)
Drones
Satellite Connectivity
Cloud Computing & Edge Computing
Generative AI
Quantum Technologies
Hydroelectric Power
Cybersecurity Technologies
Blockchain Technologies
Computer Vision, Language Processing & Object Recognition
Battery
Hydrogen Fuel Cells
EVs & Charging Infrastructure
Water Management
Wind Energy
Smart Grid
Robotics
Environmental Engineering
Sustainable Agriculture
Geothermal Energy
Recycling
Carbon Capture & Storage
Energy Efficiency
Green Consumer Goods
Solar Energy
Heat Pumps
Other
Waste Management
Biomaterials
Pollution Control
Biomass Energy
High−Performance Computing (HPC)
Semiconductors/Chips
Green Building
Software Engineering & System Development
Artificial Intelligence
Biofuels
Nuclear Energy

0 10 20 30 40
0.5 1.0 2.0

Pearson (log RCA) = 0.50 (p=0.001) Spearman = 0.54 (p=0.000) n = 39

China (CN)'s reliance on Greater Manchester (UKD3) (RCA)

Quantum Technologies
Mobile Networks (5G/6G)
Drones
Satellite Connectivity
Internet of Things (IoT)
Wind Energy
Carbon Capture & Storage
Cloud Computing & Edge Computing
Blockchain Technologies
Cybersecurity Technologies
Geothermal Energy
EVs & Charging Infrastructure
Robotics
Smart Grid
Environmental Engineering
Energy Efficiency
Recycling
Other
Green Building
Artificial Intelligence
Nuclear Energy
Heat Pumps
Hydrogen Fuel Cells
Green Consumer Goods
Battery
Biomaterials
Computer Vision, Language Processing & Object Recognition
Semiconductors/Chips
Waste Management
Generative AI
Pollution Control
Water Management
Sustainable Agriculture
Biofuels
Biomass Energy

0 10 20 30 40
0.2 0.5 1.0 2.0 5.0

Relatedness density of Greater Manchester (UKD3)

Pearson (log RCA) = 0.15 (p=0.367) Spearman = 0.18 (p=0.265) n = 39

Greater Manchester (UKD3)'s reliance on China (CN) (RCA)

Mobile Networks (5G/6G)
Internet of Things (IoT)
Satellite Connectivity
Drones
Cloud Computing & Edge Computing
Generative AI
Quantum Technologies
Hydroelectric Power
Blockchain Technologies
Cybersecurity Technologies
Computer Vision, Language Processing & Object Recognition
Battery
Hydrogen Fuel Cells
EVs & Charging Infrastructure
Robotics
Smart Grid
Environmental Engineering
Sustainable Agriculture
Geothermal Energy
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Energy Efficiency
Recycling
Heat Pumps
Carbon Capture & Storage
Solar Energy
Waste Management
Biomaterials
Green Building
High−Performance Computing (HPC)
Biomass Energy
Semiconductors/Chips
Pollution Control
Artificial Intelligence
Software Engineering & System Development
Biofuels
Nuclear Energy

50 60 70 80 90 100
0.5 1.0 2.0

Relatedness density of China (CN)

Relatedness density of Greater Manchester (UKD3)

Pearson (log RCA) = −0.05 (p=0.776) Spearman = −0.05 (p=0.764) n = 39

China (CN)'s reliance on Greater Manchester (UKD3) (RCA)

Quantum Technologies
Mobile Networks (5G/6G)
Drones
Satellite Connectivity
Wind Energy
Internet of Things (IoT)
Carbon Capture & Storage
Cloud Computing & Edge Computing
Blockchain Technologies
Cybersecurity Technologies
Geothermal Energy
Nuclear Energy
Environmental Engineering
Robotics
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Hydroelectric Power
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Generative AI
Pollution Control
Water Management
Sustainable Agriculture
Biofuels
Biomass Energy

50 60 70 80 90 100
0.2 0.5 1.0 2.0 5.0

Relatedness density of China (CN)